1/29/26

then, to undo this action, we must first take off our shoes and then take off our socks (i.e. $\underset{\uparrow}{b^{-1}}\underset{\uparrow}{a^{-1}}$).

Proof: We compute $b^{-1}a^{-1}\cdot ab = b^{-1}ea$
$= b^{-1}b = e$. $ab\,b^{-1}a^{-1} = aea^{-1} = aa^{-1} = e$.
Thus $b^{-1}a^{-1}$ is an inverse to ab.
However, since the inverse to any element is unique, then $b^{-1}a^{-1} = (ab)^{-1}$. ∎

– Abstract Algebra 2/3/26

Recall: In ≈ Socks & Shoes

Claim: $(ab)^{-1} = b^{-1}a^{-1}$.

Previous proof: We computed $(b^{-1}a^{-1})(ab)$
$= b^{-1}a^{-1}ab = b^{-1}(a^{-1}a)b = b^{-1}eb = b^{-1}b$
$= e$

Our math debt: If $cd = e$, then $c = d^{-1}$ (by context)

Proof: Suppose $cd = e$, where $c, d \in G$ (a group). Since $d \in G$, $\exists d^{-1} \in G$ s.t. $dd^{-1} = e$.

So, group appl. both sides by d^{-1}, on the right, we get $(cd)d^{-1} = \underbrace{(e)d^{-1}}$.
By associativity, $(cd)d^{-1} = c(dd^{-1}) = ce$, by "in-groes" (legacy of group) by transitivity, $ce = ed^{-1}$, (mapping group) and by "identity" $c = d^{-1}$. ∎

Reminder: In general, our proofs should be in complete sentences.

Reminder: no office hours tomorrow, instead, office hours this Friday over Zoom at 8–10 AM.

Recall: D_3 = The group of symmetries of a Δ
$= \langle r, s : r^3 = s^2 = id.$ and $rs = sr^2 \rangle$

D_n = the group of symmetries of a regular n-gon
= the dihedral group on n-vertices
$= \langle r, s : r^n = s^2 = e$ & $rs = sr^{n-1} \rangle$
↑ ↑ vertical flip
(rotation by $\frac{360}{n}$ clockwise)

Done with Chapter 2 2/3/26

Chap 3: Finite Groups & Subgroups

Def^n: The number of elements in a group G is called its order. We denote it $|G|$.
E.g. $|\mathbb{Z}| = \infty$, $|\mathbb{Z}/2\mathbb{Z}| = 2$,
$|D_3| = 6$, $|D_4| = 8$, $|\mathbb{Z}/17\mathbb{Z}| = 17$.
If a is an element of a group G, we define the order of a to be the least positive integer n such that $\underbrace{a \times a \cdots \times a}_{n\text{-times}} = a^n = e$. We denote this as $|a| = n$
group appl. n-times
If no such n exists, we say the order of a is infinite and we write $|a| = \infty$.

Note: For $n \in \mathbb{N}^+$, def^n $a^n = \underbrace{a \cdot a \cdots a}_{n\text{-times}}$

Def^n: $a^0 = e$ a inverses

And $a^{-n} = \underbrace{a^{-1} \cdot a^{-1} \cdots a^{-1}}_{n\text{-times}}$

group appl. group appl. unlimited

Fact: $a^n \cdot a^m = a^{n+m}$ and $(a^n)^m = a^{nm}$

Ex. Let $r \in D_3$. $|r| = 3$.
What about $|rs|$?
$(rs)^2 = rsrs = sr^2rs = s^2r^2s = ses = s^2 = e$
Thus, $|rs| = 2$.
Consider $1 \in (\mathbb{Z}, +)$. We normally assume the group appl. is addition, unless we use "x" as a superscript.
$|1| = \infty$
$(\mathbb{Z}, \times) = \{1\}$ for fact.

$(\mathbb{Q}, +)$. Consider $\frac{1}{3} \in \mathbb{Q}$. $|\frac{1}{3}| = \infty$
which means $\underbrace{\frac{1}{3} + \cdots + \frac{1}{3}}_{n\text{-times}} \neq \theta$
What is $|\bar{2}|$ where $\bar{2} \in \mathbb{Z}/6\mathbb{Z}$?
Well, $\bar{2} \neq \theta$, so $\bar{2} + \bar{2} = \bar{4} \neq \theta$.
$\bar{2} + \bar{2} + \bar{2} = \bar{6} = \theta$ ✓ So, $|\bar{2}| = 3$

(Josh's) Conjecture: 2/3/26

If $\gcd(n, m) = 1$, then $|\bar{a}|$ in $\mathbb{Z}/m\mathbb{Z}$ is m & $|\bar{a}| = m$.

Def^n: A subgroup H of a group G is a subset of G that is a group under the opperation of G.
We denote this: $H \leq G$.
{in this context, this is not "less than or equal to"}

Why do we have order of an element and order of a group?
The order of an element $a \in G$, is the order of the smallest subgroup containing a.
this is the smallest subgroup containing a

Consider $H = \{r, s^2, e\} \subset D_3$.
Claim: $H \leq D_3$. Is H associative? Yes, because D_3 is. Have identity?

	e	r	r^2
e	e	r	r^2
r	r	r^2	e
r^2	r^2	e	r

we see closure ✓ and inverses ✓

There is also symmetry along the diagonal, so H is Abelian.

Ex. Consider $6 \in (\mathbb{Z}, +)$.
What is the smallest subgroup of $\mathbb{Z}$ containing 6? Let's call this the subgroup generated by 6, $\langle 6 \rangle$ = $\{\ldots, -24, -18, -12, -6, \theta, 6, 12, 18, 24, 30, \ldots\}$ (inverses need to be here too) (id needs to be there) from group appl.
$= 6\mathbb{Z}$ = the multiples of 6
$|\langle 6 \rangle| = \infty = |\mathbb{Z}|$

– Abstract Algebra 2/3/26

Reminder: Professor's office hours will be tomorrow Friday, 8 AM – 10 AM, and also next Friday.
HW is due Sunday, this week and next.